ARtPUT: Autonomous Rover to Pick Up Trash

Progress Report 7: Hardware Assembly and Component Integration

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Materials and Methods

Materials:

- DFRobot Devastator Tank Mobile Robot Platform serving as the tracked mobile base
- Raspberry Pi 4 Model B as the primary onboard computer for vision, navigation, and logging
- Arduino UNO for low-level motor, servo, and sensor control
- USB RGB camera with depth-sensing capability for object detection and grasp planning
- Ultrasonic sensors for short-range obstacle detection
- NEO-6M GPS module for position logging and route tracking
- Digital compass (magnetometer) for heading correction
- Infrared sensor for bin fill-level detection
- Two-finger robotic gripper actuated by a servo motor
- 12V DC suction pump with relay module
- Suction-assisted collection bin with approximately 500–1000 mL capacity
- L298N motor drivers for drive motors
- Buck converters for voltage regulation
- Two 11.1 V Li-Po batteries to power onboard systems
- Wireless telemetry and onboard data logging storage
- Cones, measuring tape, stopwatch, and field notebook for controlled outdoor testing

Procedure:

1. Dataset & training: To train an object-detection model (YOLO/SSD-style), gather and label a significant number of photos of the target litter types in the target park. Use artificial occlusions to enhance.
2. Software stack: Put in place a perception pipeline (object detection on RGB-D + depth clustering), SLAM-based localization (Lidar + odometry), and a manipulation controller that translates detected object pose to a grasp technique. We'll test two modes of perception manipulation: Vision-only bounding-box grasping (A); vision plus depth-point-cloud grasp planning (B).
3. Three 10 m × 10 m test plots in a public park with low pedestrian traffic are chosen for the test. A fixed number (N = 20) of different litter items are placed in each plot at random locations (pre-approved with park authority). Mark the start and goal points.
4. Trials: For each mode (A, B) run 10 trials per plot (total 60 runs), each trial starting from the same location, time-limited to 15 minutes or until bin full. Rotate item arrangements between trials. Record video, sensor logs, and manual observer notes.
5. Baseline: Include 10 human-assisted pickup runs (human uses same route, picks items by hand) for performance comparison.

Data Collection Methods:

- Automated logs: timestamps of detections, attempted grasps, success/failure flags, battery consumption, distance travelled.
- Manual annotations: items missed, misclassifications, pedestrian interactions, environmental notes (wetness, wind).

- Post-run bin content check to count unique items collected.

Safety Concerns:

- The use of electronics and batteries in our project presents a risk of burns and electrical shock.
- The gears in our system may catch onto our limbs and clothing, potentially crushing our limbs.
- The rover may move erratically and hit someone, causing a harmful impact.
- We may drop heavy objects on our feet.

Risk Mitigation:

- We will always wear safety glasses, gloves, and proper footwear while working on our project.
- We will not wear baggy clothes that could potentially get caught in gears.
- There will always be a supervising adult while we are working on our project.
- The rover will always be fully powered off when working on it.

Data Analysis

Since our previous progress report, we have made very significant progress on the hardware assembly of ARtPUT. Despite significant delays caused by late component deliveries and a demanding testing schedule in recent weeks, we fully assembled the Devastator tank chassis, installed both rubber tank treads, and mounted both drive motors securely within the frame. Both 11.1V LiPo batteries were mounted inside the chassis, and two LM2596 buck converters were precisely calibrated — one to 5.10V for the Raspberry Pi 4, and one to 5.00V for the Arduino Uno — providing clean, regulated power to all logic electronics while keeping the motor drive circuit on a separate power domain.

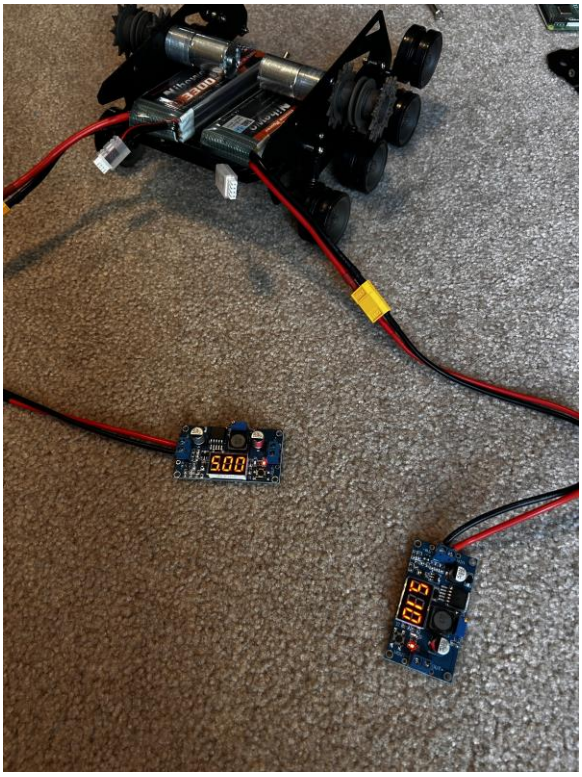

Figure 1: This is a bench test confirming successful operation of both LM2596 buck converters, with outputs calibrated to 5.10V for the Raspberry Pi 4 and 5.00V for the Arduino Uno.

The L298N dual H-bridge motor driver was mounted on the underside of the top deck plate for weather protection and wired to both drive motors. Its control signal pins were connected to the Arduino Uno via jumper wires, enabling full software-controlled speed and direction for both motors. The Raspberry Pi 4 and Arduino Uno were mounted on the top deck, powered on, and confirmed fully operational. Both computing boards will communicate via a USB serial connection, with the Pi handling high-level autonomous decision making and the Arduino managing low-level motor and sensor control.

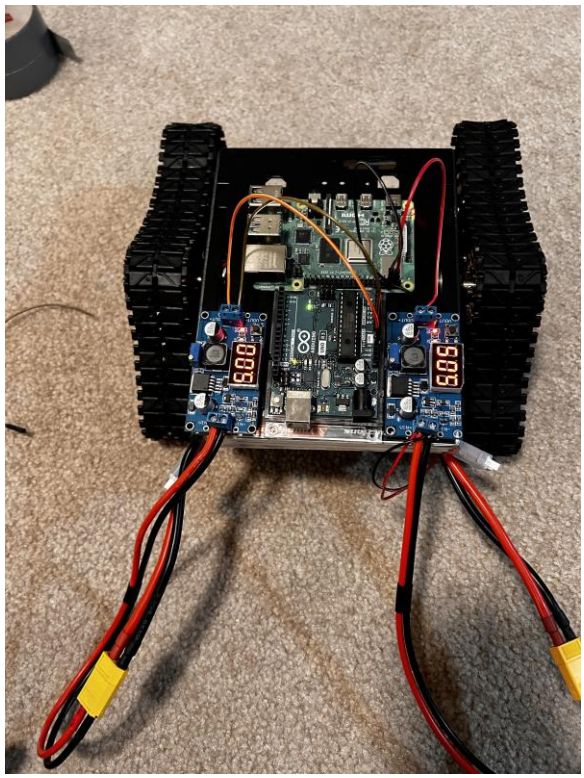

Figure 2: This is a top deck view of ARtPUT with the Raspberry Pi 4 and Arduino Uno mounted and connected, confirming both computing boards are operational.

Three HC-SR04 ultrasonic sensors were mounted on the front face of the robot in a fan configuration for wide-angle obstacle detection, and the Sharp GP2Y0A21 infrared distance sensor was mounted at the front-lower position to detect objects within the gripper's operating

range. Sensor wiring will be finalized tomorrow using a solderless breadboard. Remaining tasks include gripper arm assembly, flashing the Raspberry Pi OS, and writing the autonomous control software, all of which are scheduled for Wednesday. The robot is on track for a full autonomous trash pickup test run by Thursday, with data to be ready by Friday.

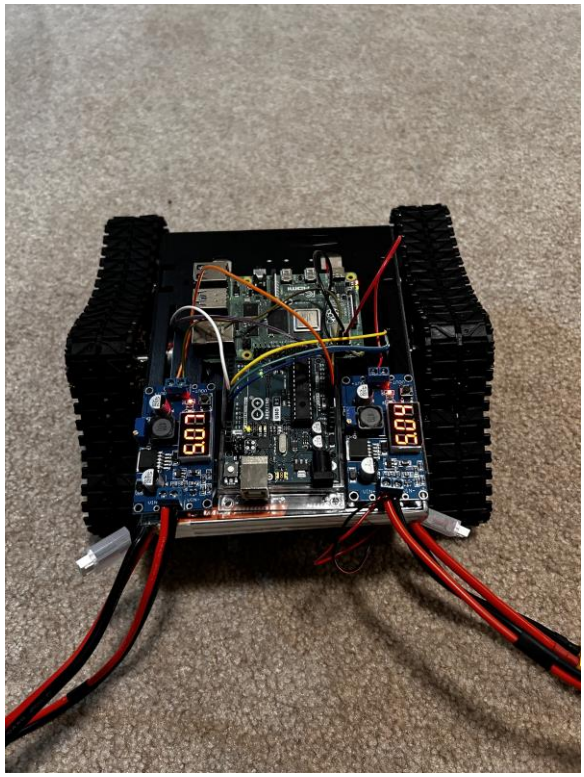

Figure 3: This is a fully wired top deck view of ARtPUT with all onboard components connected, including the motor driver, computing boards, and remaining electronics.

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